

BLY 111:

TOPIC: PLANT AND ANIMAL CELLS

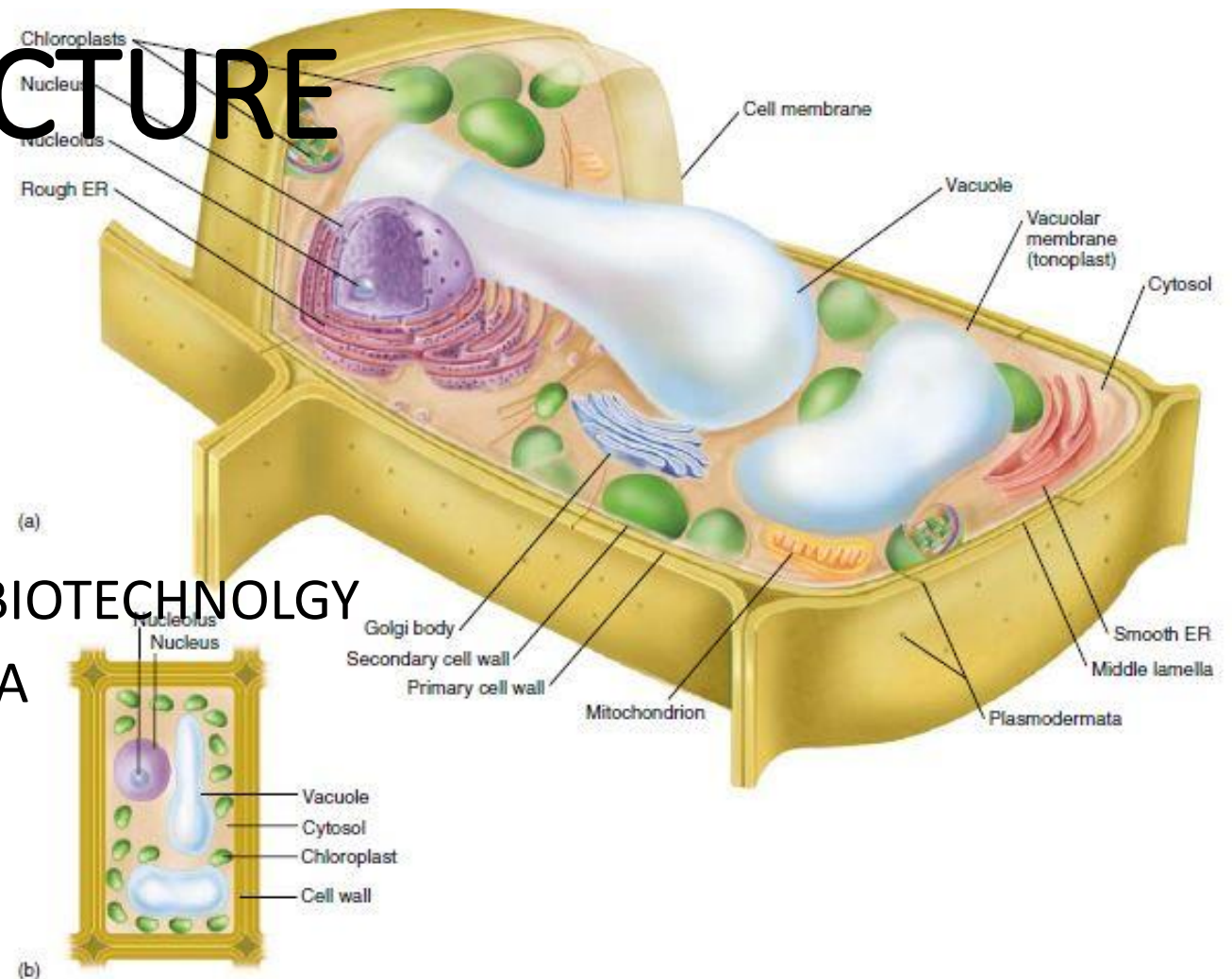
STRUCTURE

BY

DR. AKOMOLAFE G. F.

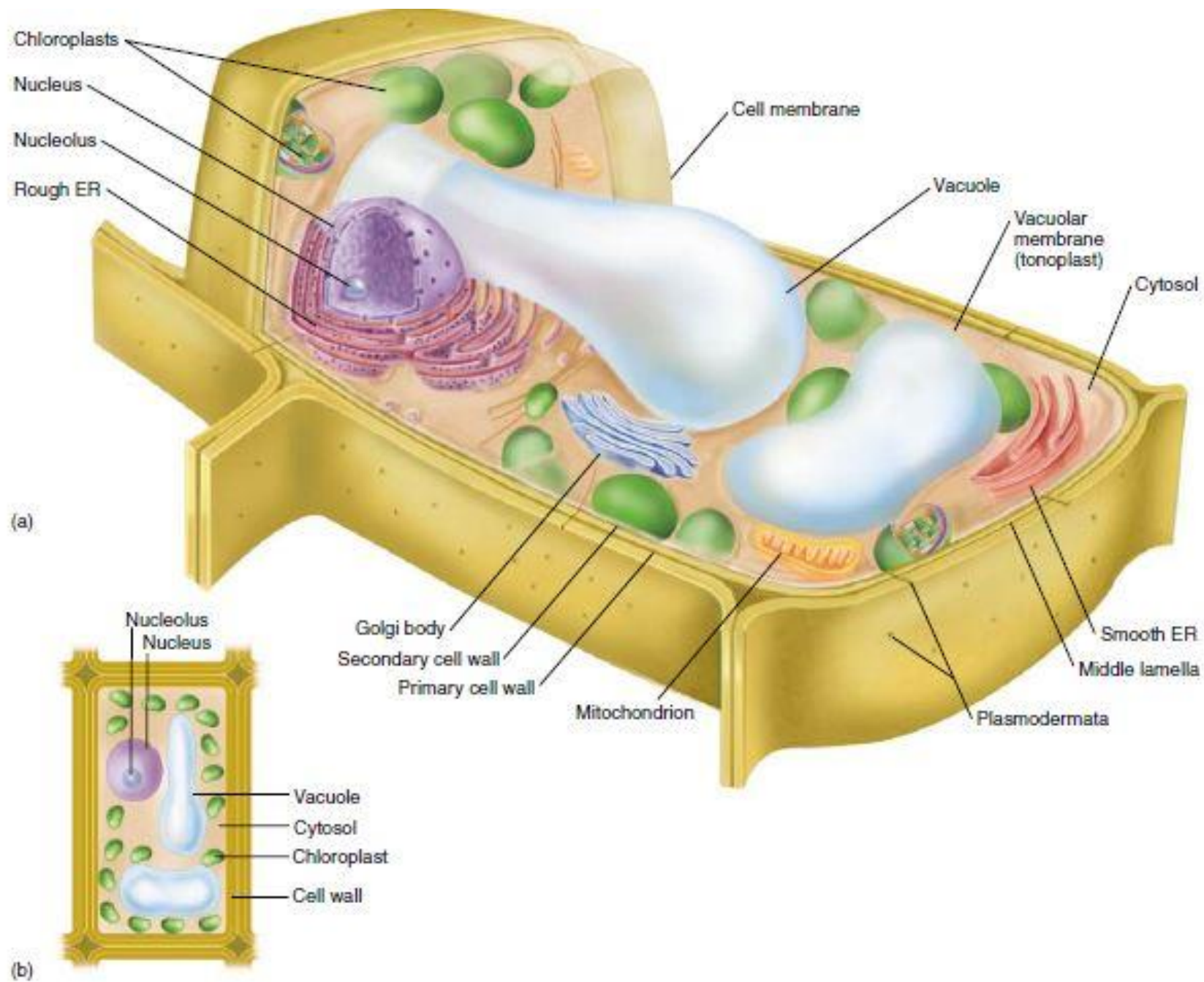
DEPARTMENT OF PLANT SCIENCE AND BIOTECHNOLOGY

FEDERAL UNIVERSITY LAFIA



INTRODUCTION

- The cell can be defined as the fundamental structural and functional unit of life
- They are the physiological and morphological unit of the organism
- Plant and animal cells are eukaryotic (true nucleus) and have many of the structures found in animal cells
- Plant cells differ from animal cells in lacking centrioles and intermediate filaments having plastids most especially chloroplast, cell walls and large vacuoles
- The plant cells can be divided into two parts; cell wall and protoplast.



CELL CONTENTS

- Immediately within the cell wall, the cell contents are delimited by a membrane, the plasma membrane (plasmalemma).

Endoplasmic Reticulum:

- it is an elaborate system of membrane within the cytoplasm
- It consists of double membrane that encloses spaces or cisternae.
- The membrane of endoplasmic reticulum is of lipoprotein nature and is connected with the nuclear envelope and also extends to the margin of the cell
- The ER occurs in two forms: rough and smooth.
- The rough one is rendered rough by the occurrence of numerous small particles of ribonucleoproteins on its outer surface

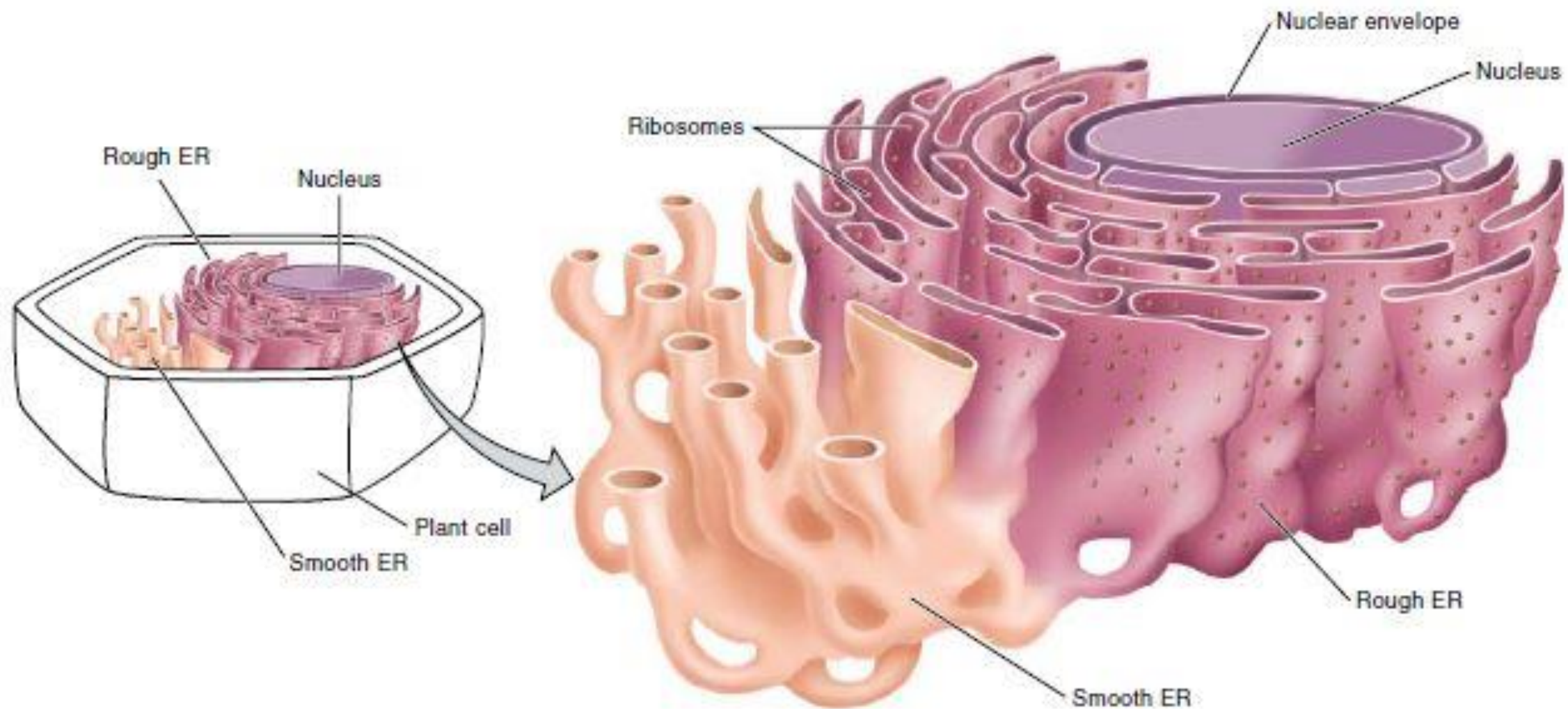


FIGURE 4.4

The three-dimensional system of flattened tubes and sacs of the ER. Ribosomes are the site of protein synthesis.

Dictyosomes

- the Golgi apparatus of plant cells consist of a system of dictyosomes occurring throughout the cytoplasm.
- A dictyosome consists of a small stack (2 to 20) of smooth double membranes enclosing cisternae which are often dilated at the ends.
- They are associated at the edges with a number of small vesicles

Microtubules and Microfilaments:

- microtubules are small elongated structures and are present in peripheral regions of the cytoplasm
- and in some other part of the cell.
- They are made of a protein called tubulin which in it's polymerise form is visible as microtubules.

Nucleus:

- in meristematic cells, the nucleus occupies a considerable proportion of the volume of the cell (sometimes as much as $\frac{2}{3}$ or $\frac{3}{4}$).
- This problem become less during differentiation.
- The nucleus is bounded by a double membrane, the nuclear envelope which has pores in it at regular interval.
- The nuclear envelope is connected with the endoplasmic reticulum.
- One or more nucleoli may be present.
- These are not bounded by membrane.
- Chromosomes are present and consist of nucleoprotein comprising DNA and protein including histones.
- The nucleolus on the other hand contains RNA and protein.

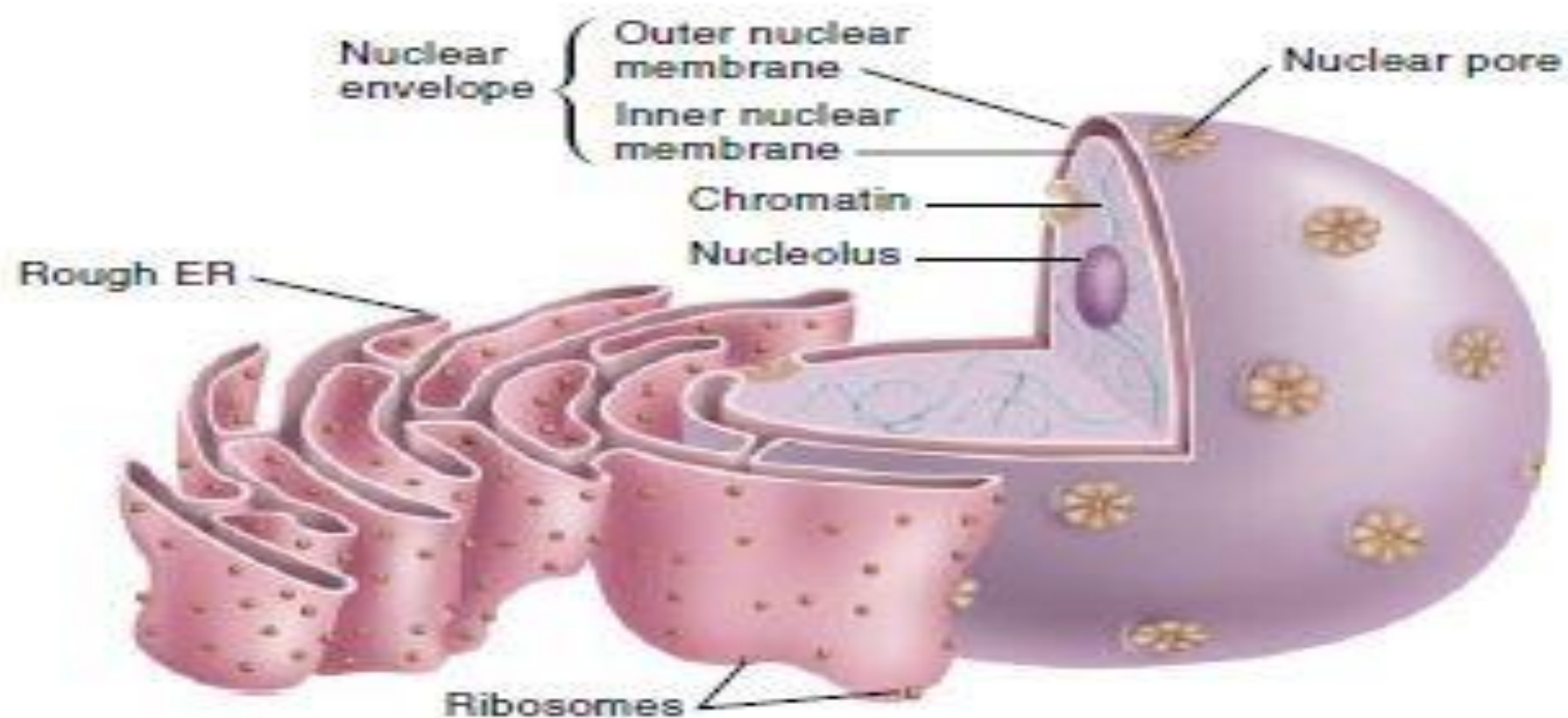


FIGURE 4.3

The nuclear envelope has pores that link the cytoplasm with the inside of the nucleus. The outer nuclear membrane is continuous with the ER. Chromatin is DNA plus protein that together constitute chromosomes.

Ribosomes:

- they are small, more or less spherical organelles which occur free in the cytoplasm on the outside of the ER and in the nucleus, chloroplast and mitochondria.
- Clusters of ribosomes are known as Polyribosomes or Polysomes

Spherosomes:

- the majority of spherosomes consist of lipid droplets which are not bounded by a membrane.
- Their frequency of occurrence varies in different cells.

Microbodies:

- they are small bodies and are essentially of two types: glyoxysomes and peroxisomes.
- Both have a single bounding membrane and are wide spread in plant cells.
- They differ mainly in the enzymes present in them.
- Peroxisomes occurs in leaves often contiguous with chloroplast envelope.
- Glyoxysomes occur in the storage tissues of fatty seeds

Mitochondria:

- they are visible as small rods.
- The electron microscope reveals that they are bounded by a double membrane.
- Extensions of the inner membrane form tubular projections or microvilli or plate-like folds called cristae.
- The matrix of the mitochondrion is mainly made up of protein.
- Ribosomes are present and slightly smaller than those in the cytoplasm.

Plastids:

- this may be divided into 2 types: Pigmented and non-pigmented.
- Pigmented plastids consist of Chloroplast which are green in colour containing the chlorophyll pigments
- and Chromoplast which are usually yellow, orange or red and contain carotene pigment.
- Non-pigmented plastids or Leucoplasts include the Amyloplast
- and Elaioplasts.
- These plastids are all derived from very small bodies, the Pro-Plastids.
- If the development of a pro-plastid into a chloroplast is interrupted by lack of light, an Etioplast is formed.

Ergastic Substances:

- these are formed as a result of metabolic activities in the cell.
- Some of these are waste products. Others are stored food materials.
- Starch: the carbohydrate starch is composed of long-chain molecules which are symmetrically spaced and consequently confer some crystalline properties on starch grains.
- Starch grains may be found in all parenchymatous tissues but especially in storage organs such as tubers, corms, rhizomes, endosperm or cotyledons of seeds.
- Starch is synthesized by the chloroplast.
- Proteins: it sometimes occurs in reserved materials.
- Fats and Oils: they are widely distributed in plant cells.
- They are chemically similar. Fats usually being solid and oil liquids. Waxes, Cutin and Suberin which occur in and upon the cell wall are also fatty.
- Crystals: Crystals occur in the cells of many plants most consist of salts of calcium

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